Research Vision — Reliable, Data-Efficient AI for Geospatial Intelligence and Multisensor Systems

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Vision and research foundations

My research vision is to develop reliable, data-efficient and interpretable artificial intelligence for geospatial perception and decisions under real-world uncertainty. Remote-sensing systems must operate across geographic regions, seasons, viewpoints, sensor resolutions, and incomplete observations. In national-security and public-interest applications, errors can have substantial consequences; accuracy on a single benchmark is not enough. I propose a research program integrating machine learning, computer vision, multisensor data fusion, uncertainty-aware decision support, and resource-efficient computation, with explicit evaluation of when and why AI systems fail.

I approach this opportunity from an established foundation in AI methodology and intelligent sensing rather than from an existing satellite-imagery research portfolio. My 2025 PhD in Electrical and Computer Engineering at the University of Oklahoma focused on AI/ML for wireless networks, particularly data scarcity, distribution shift, generative augmentation, reliable network prediction and adaptive optimization. Peer-reviewed research in Computer Communications, Sensors, IEEE Network, IEEE PIMRC and IEEE VTC has explored transferable learning, AI-enabled anomaly detection, and learning-driven recovery. In parallel, smartwatch-based human-activity research has examined deep learning on noisy sensor time series and generalization across users. Industrial AI work with AT&T, AT&T Labs, Futurewei and Impac Exploration Services has emphasized operational data, diagnostic errors, validation, and deployment constraints. Together, these experiences provide a methodological starting point for original geospatial AI research in partnership with domain experts.

Theme 1: Robust visual learning under geographic and sensor shift

Research question. How can feature detection and scene understanding remain dependable when labeled imagery is limited and operational conditions differ from the training distribution?

I will study transfer learning, domain adaptation, data-efficient generative augmentation, and uncertainty calibration for object detection, segmentation and change detection in remote-sensing imagery. A core hypothesis is that domain-aware augmentation and selective adaptation can improve out-of-distribution reliability more consistently than indiscriminate synthetic-data expansion. Initial experiments would use openly licensed datasets such as SpaceNet building/road imagery and xView overhead object-detection benchmarks, complemented by additional public imagery where licensing permits. Baselines would include conventional supervised vision models and contemporary detection/segmentation architectures, then add carefully controlled pretraining or augmentation strategies.

Evaluation will separate images by region, collection conditions, sensor type and time where metadata allow. Beyond aggregate accuracy, I will measure per-class precision/recall, rare-target performance, confidence calibration, false alarms, computational cost, and worst-group errors. Experimental protocols will distinguish genuine cross-region generalization from random train/test splits that leak spatial context. The first methodological outcome will be a reproducible benchmark suite and a study of domain-shift failure modes; subsequent work will develop and test new data-efficient adaptation methods.

Theme 2: Multisensor and spatiotemporal fusion

Research question. When does combining complementary observations produce a more robust result than relying on a single sensor, especially if modalities are missing, delayed, noisy or misaligned?

My prior work in wireless localization, signal analysis, heterogeneous network measurements, and wearable time-series learning motivates a shift toward multimodal geospatial representation learning. Proposed experiments would investigate fusion of optical imagery, radar-derived observations, geographic or positional metadata, and temporal measurements in

appropriately selected public datasets. Candidate methods include late fusion, attention-based feature fusion, uncertainty-weighted integration, modality dropout, and representation alignment across sensor types. Where optical/radar registration is part of the problem, the system will explicitly model alignment uncertainty instead of silently assuming perfect correspondence.

One application would be change detection and infrastructure monitoring, where temporal imagery and contextual measurements can help distinguish meaningful changes from illumination, weather or sensor artifacts. Another would be reliable small-object detection using multiple observations. The objective is not simply to add modalities, but to quantify when fusion helps, when it fails, and how the system signals unreliable evidence. Research artifacts would include openly reproducible pipelines, calibrated uncertainty estimates, ablation studies and student-led publications.

Theme 3: Trustworthy, efficient AI for analyst-supported decisions

Research question. How can resource-constrained inference and human oversight improve the operational usefulness of AI-generated geospatial alerts?

I will connect perception models with uncertainty-aware decision support, risk-sensitive prioritization and efficient deployment. Building on reinforcement learning, network optimization and operational AI diagnostics, I will investigate how detection confidence, data quality, expected consequences of errors and inference cost can jointly guide automated triage. Technical methods may include selective prediction, confidence- and cost-aware ranking, compact/edge models, adaptive processing and human-in-the-loop review. Research on high-performance computing will focus on scalable preprocessing, experiment orchestration and efficient training/evaluation of large imagery collections in collaboration with campus computing specialists.

This direction will preserve human authority in consequential decisions. I will treat fairness, uncertainty, provenance, data rights, and false-positive burden as explicit engineering criteria. Initial demonstrations will use non-sensitive open datasets, including disaster response or infrastructure monitoring as dual-use testbeds. Access to controlled data, defense programs or sensitive tasks would occur only through authorized partnerships, institutional review, and applicable security and export-control procedures. I would not assume that every federal or defense project is available to all personnel.

Research roadmap and external funding

First 12 months: Establish reproducible visual-learning baselines; identify target geospatial problems with CGI collaborators; recruit undergraduate and graduate research students; publish an initial benchmark/evaluation study; and develop one collaborative seed or external proposal. Emphasis will be on experimental rigor, public data and a precise problem definition, not unverified performance targets.

Years 2–3: Develop at least two integrated methodological projects: (1) domain-adaptive and calibrated geospatial detection and (2) incomplete-modality fusion for change detection or spatial monitoring. Submit journal/conference manuscripts, establish shared code and data-governance practices, and prepare proposals through programs where eligibility and solicitation requirements match the research. Potential sponsor categories include NSF computing/AI programs and relevant federal geospatial, defense and public-sector research opportunities pursued with experienced campus partners.

Years 4–5: Integrate reliable perception, multimodal fusion and resource-aware decision-support components into a coherent research testbed. Pursue larger interdisciplinary awards and build collaborations with faculty in sensing, computer vision, signal processing, geospatial informatics, optimization, and high-performance computing. I would engage CGI for problem definition and mission-relevant evaluation, and Mizzou AERI for responsible AI partnerships and computational resources, subject to collaboration opportunities.

I have participated in research projects and proposal-development activities, but do not claim a prior externally funded award as principal investigator. My funding strategy will therefore begin with focused, credible proposals and experienced collaborators, then grow toward an independent externally funded research program.

Graduate education and wider impact

The proposed work aligns with the University of Missouri's expanding MS and PhD programs in Artificial Intelligence. I would contribute to teaching machine learning, deep learning, data science, reliable AI, intelligent sensing and application-focused computer vision as appropriate to curriculum needs and preparation. Students could begin with carefully scoped tasks such as cross-region benchmark design, missing-modality ablation studies, confidence calibration and error analysis, then progress to original methods and interdisciplinary doctoral projects. Undergraduate research assistants would participate in reproducible experiments, mentoring and technical communication.

The broader impact lies in making AI systems more trustworthy in unfamiliar environments, not just more accurate on familiar datasets. Such methods support mission-relevant geospatial research while retaining utility for disaster response, environmental monitoring, infrastructure assessment and other socially valuable applications. The University of Missouri's combination of EECS, the Center for Geospatial Intelligence, and Mizzou AERI offers a promising environment for this interdisciplinary, method-driven research agenda.